

TIANMU SHAN, China

WMO: 584450

Lat: 30.3494N

Lon: 119.4237E

Elev: 1494

StdP: 84.62

Time Zone: 8.00 (E08)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-9.6	-30.5	0.3	-2.2	-25.8	0.4	-2.4	18.0	-5.9	16.4	-2.5	8.4	340	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	25.3	21.2	24.1	20.6	22.9	20.0	21.9	24.3	21.2	23.3	20.6	22.3	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	19.0	23.5	20.6	18.3	22.6	20.0	17.7	21.7	72.5	24.4	69.8	23.5	67.2	22.4	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
15.9	13.6	11.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-14.9	27.3	1.9	1.4	-16.3	28.3	-17.4	29.1	-18.5	29.9	-19.9	30.9		
			-15.8	23.1	1.7	0.6	-17.0	23.5	-18.0	23.8	-19.0	24.2	-20.2	24.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-1.8	-0.5	3.5	9.3	13.5	16.8	19.9	19.1	14.8	10.2	5.8	1.5
	DBStd	8.30	4.35	4.90	4.74	4.26	3.60	2.41	2.53	2.14	3.16	3.66	5.07	5.02
	HDD10.0	1390	364	295	208	62	11	0	0	0	2	42	141	265
	HDD18.3	3386	623	527	461	272	153	56	15	17	112	252	376	523
	CDD10.0	1172	0	1	6	40	119	204	308	283	146	49	14	1
	CDD18.3	126	0	0	0	0	3	11	64	42	6	0	0	0
	CDH23.3	180	0	0	0	0	4	6	121	44	5	0	0	0
	CDH26.7	6	0	0	0	0	0	0	4	1	0	0	0	0
Wind	WSAvg	5.5	5.7	5.4	5.6	5.6	5.6	5.4	5.5	5.5	5.4	5.1	5.4	5.4
Precipitation	PrecAvg	1690	55	81	124	149	197	253	200	223	190	102	72	46
	PrecMax	2085	121	219	263	241	390	513	395	422	491	320	207	132
	PrecMin	1126	0	17	45	68	74	85	34	35	52	1	2	0
	PrecStd	265	30	44	52	45	73	103	93	103	100	81	50	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	13.6	15.8	21.4	23.7	24.4	27.4	26.0	24.9	20.6	17.3	13.4
		MCWB	4.9	9.0	12.3	14.0	17.4	19.2	22.1	21.7	21.2	15.5	12.0	6.1
	2%	DB	7.6	10.7	13.7	18.8	21.5	22.6	25.9	24.5	22.4	18.3	15.3	11.2
		MCWB	4.6	7.8	11.4	14.1	16.5	19.6	21.7	21.2	19.1	13.7	10.9	5.3
	5%	DB	6.3	9.0	12.1	16.5	20.0	21.1	24.8	23.3	20.6	16.5	13.7	9.6
		MCWB	3.9	7.1	10.4	13.5	15.6	18.5	21.1	20.4	18.5	13.3	9.8	5.4
	10%	DB	4.6	6.8	10.6	15.0	18.3	19.9	23.5	22.0	19.0	14.8	12.2	7.9
		MCWB	1.9	4.6	9.1	12.8	14.8	18.0	20.6	19.7	17.9	12.2	8.9	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	10.6	14.1	17.1	19.8	21.3	23.1	22.5	21.8	17.6	15.2	10.0
		MCDB	8.2	12.0	14.6	18.5	21.7	22.7	26.0	24.9	24.6	19.5	15.9	10.3
	2%	WB	6.5	9.1	12.3	15.6	18.2	20.3	22.2	21.6	20.2	15.9	13.3	8.2
		MCDB	7.1	9.9	12.9	16.5	19.8	21.6	24.8	23.7	21.8	16.8	14.1	9.1
	5%	WB	5.0	7.5	11.1	14.6	17.2	19.5	21.6	20.9	19.2	14.3	11.9	6.8
		MCDB	5.4	8.6	11.8	15.7	18.4	20.4	24.0	22.6	20.1	15.2	12.7	8.6
	10%	WB	2.7	5.1	9.6	13.4	16.3	18.8	20.9	20.2	18.2	12.8	10.1	5.1
		MCDB	3.6	6.4	10.3	14.6	17.4	19.5	23.1	21.5	18.8	14.0	10.9	7.4
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.8	6.0	6.1	5.6	4.4	4.5	4.2	4.4	5.4	6.1	6.2
		MCDBR	7.1	7.9	7.6	7.9	7.4	6.0	6.8	6.2	6.2	7.0	7.0	7.5
		MCWBR	5.5	5.8	6.0	5.9	5.7	4.8	4.4	4.4	5.3	5.5	5.5	5.8
	5% WB	MCDBR	5.6	6.6	6.3	6.2	5.9	4.6	5.9	5.4	5.0	5.7	6.1	6.5
		MCWBR	5.7	5.6	5.6	5.2	4.8	3.7	4.2	4.1	4.3	4.6	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.435	0.470	0.500	0.504	0.483	0.509	0.482	0.500	0.489	0.462	0.438	0.416	
	taud	1.940	1.880	1.842	1.891	2.002	1.987	2.116	2.039	2.005	1.980	1.980	1.989	
	Ebn at Noon	761	775	785	801	818	794	814	795	787	774	753	757	
	Edh at Noon	156	181	201	199	179	181	158	169	169	162	148	142	
All-Sky Solar Radiation	RadAvg	2.12	2.44	3.43	4.05	4.40	3.98	5.06	4.47	3.64	3.23	2.43	2.17	
	RadStd	0.37	0.60	0.30	0.50	0.53	0.60	0.62	0.60	0.37	0.52	0.37	0.42	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+51

Nomenclature: See separate page